

Enhanced CLAP Representations with Neighborhood Refinement for Heterogeneous Audio Classification

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Abstract—Heterogeneous audio classification under the Broad Sound Taxonomy (BST) requires accurate second-level predictions while maintaining consistency with the top-level taxonomy structure. CLAP, a pre-trained model based on contrastive audio-text learning, has been adopted for heterogeneous audio classification due to its strong capability in learning high-level semantic representations. However, its emphasis on semantic alignment limits its ability to capture fine-grained acoustic variations, thereby constraining its performance in hierarchical sound classification. To address this issue, we propose an enhanced CLAP-based representation learning framework that integrates an auxiliary acoustic encoder branch and a gated text adaptation module, aiming to enrich acoustic details and improve text alignment. In addition, we introduce a K-nearest-neighbor (KNN)-based neighborhood-constrained inference strategy and construct an expanded training set to further improve prediction performance. Experimental results on the official DCASE 2026 evaluation set show that the proposed model achieves a hierarchical F1-score of 80.0%. Furthermore, an ensemble of complementary models within the proposed framework yields a final hierarchical F1-score of 81.1%, securing second place in the DCASE 2026 Challenge Task 1.

Index Terms—Heterogeneous Audio Classification, Broad Sound Taxonomy (BST), CLAP, Audio-Text Representation Learning, KNN-Based Neighborhood Refinement

1. INTRODUCTION

Heterogeneous audio classification under the Broad Sound Taxonomy (BST) aims to recognize sound events across different acoustic environments, recording conditions, and durations. The BST label space is organized hierarchically, with 23 second-level categories grouped under 5 top-level categories [1]. The hierarchical label structure of BST can be evaluated by the hierarchical F1-score (Hier. F1), which evaluates both fine-grained second-level category recognition and consistency with the corresponding top-level category [2]. Moreover, textual metadata paired with audio samples offer additional semantic cues, enabling multimodal approaches to jointly leverage acoustic and semantic representations for classification.

Existing methods for heterogeneous audio classification [3]–[9] mainly seek to learn robust representations from acoustic signals and textual metadata. Spectrogram-based neural audio classifiers are widely used for audio classification to model local time-frequency structures [3]–[6]. With the availability of textual metadata, Contrastive Language-Audio Pretraining (CLAP)-based methods [7]–[9] provide a natural solution for heterogeneous audio classification by aligning audio recordings and textual metadata in a shared semantic representation space. However, directly using CLAP embeddings has limitations for fine-grained BST classification. Since CLAP is pretrained for general audio-text alignment rather than target-category discrimination, its high-level semantic representations may be less sensitive to fine-grained acoustic variations [10], especially when second-level categories are semantically similar but acoustically

different. Meanwhile, although the CLAP text embedding provides useful semantic priors from textual metadata, it may not be calibrated to the BST label space. Therefore, how to preserve the audio-text semantic alignment of CLAP while improving fine-grained acoustic discrimination and semantic adaptation remains an important problem. Moreover, even with enhanced representations, conventional classifiers usually make predictions independently for each input and lack explicit strategies for prediction refinement using nearby training samples in the embedding space. Recent nearest-neighbor-based methods [11], [12] have shown that incorporating local neighborhood information can provide complementary evidence for prediction refinement.

Motivated by acoustic feature modeling and nearest-neighbor-based prediction refinement, we propose an enhanced CLAP classification system. Our system addresses the above challenges from three complementary aspects: CLAP-based representation enhancement, neighborhood-constrained inference refinement, and reliable data expansion. Our contributions are summarized as follows:

- We introduce an enhanced CLAP classification framework that improves audio representations through a supervised log-Short-time Fourier Transform (log-STFT) acoustic branch for learning fine-grained time-frequency cues and a gated text adaptation module for gently calibrating text semantics toward BST classification while preserving the original audio-text alignment.
- We apply K-nearest-neighbor (KNN)-based post-processing to exploit local class distributions in the embedding space, improving prediction stability for uncertain samples.
- We construct BSD-Grand, an expanded training set built from BSD10k-v1.2 [8], [13] and reliable BSD35k-CS [14] samples, to improve data diversity and support better model generalization.

2. PROPOSED METHOD

Our system is organized into four parts. Section 2.1 introduces the enhanced CLAP-based classification framework, which combines an acoustic branch with a gated text adapter for fine-grained BST classification. Section 2.2 presents the KNN-based post-processing strategy for refining uncertain predictions. The model and inference pipeline of the proposed method are illustrated in Fig. 1, while the BSD-Grand data construction and the final ensemble strategy are described separately in Section 2.3 and Section 2.4, respectively.

2.1. Enhanced CLAP Representation

Feature-specific acoustic branch: For the audio-side enhancement, we extract a log-STFT sequence from each audio clip and feed it into a Transformer-based acoustic encoder [15] to obtain an acoustic embedding, as shown in the acoustic branch of Fig. 1(a). The encoder consists of 6 Transformer encoder layers with a hidden dimension of 128 and 8 attention heads. To aggregate the frame-level outputs of the Transformer, temporal downsampling is applied to reduce the sequence length, followed by mean-max temporal pooling to obtain a 128-dimensional acoustic embedding.

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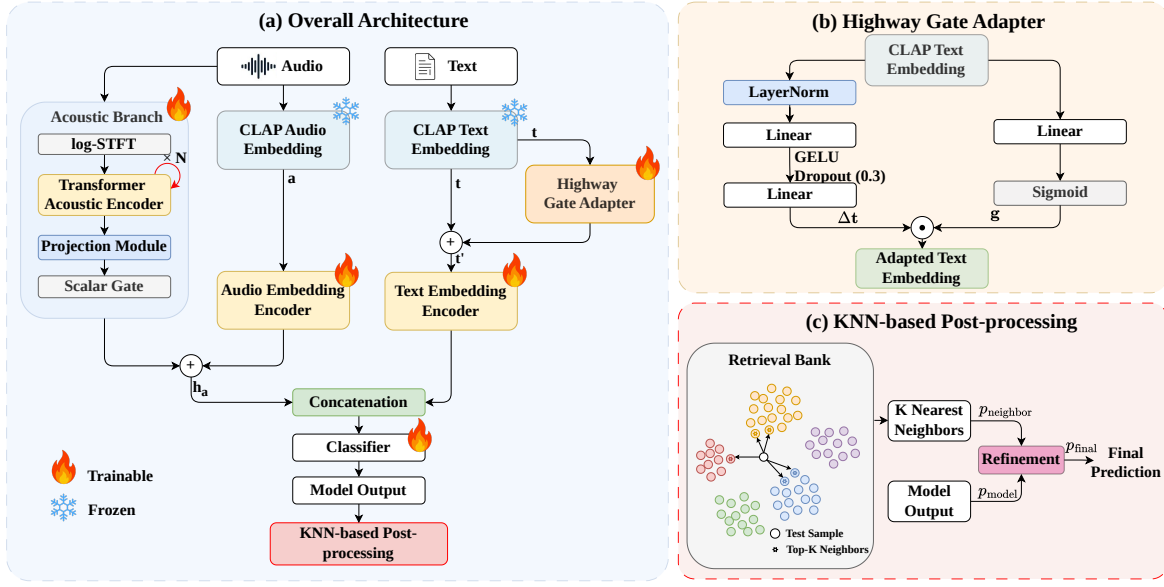


Fig. 1: Overview of the proposed framework: (a) Overall architecture; (b) Structure of the Highway Gate Adapter; (c) KNN-based post-processing for neighborhood-constrained prediction.

Since the acoustic embedding and the CLAP audio embedding lie in different feature spaces, we then use a projection module to map the 128-dimensional acoustic embedding to the 512-dimensional CLAP audio embedding space. The projected acoustic embedding is then integrated with the original CLAP audio representation through a gated residual fusion strategy [16], where a learnable scalar gate controls the contribution of the acoustic branch. Through this design, we preserve the high-level semantic information in the original CLAP audio representation while selectively incorporating fine-grained spectral and temporal cues from the log-STFT branch.

Highway Gate Adapter: For the text-side enhancement, we introduce a Highway Gate Adapter as shown in Fig. 1(b) to adapt the CLAP text embedding to BST classification.

Let $\mathbf{t}$ denotes the CLAP text embedding. The adapter first generates a residual update $\Delta \mathbf{t}$ through a bottleneck [17]. In parallel, a sample-dependent gate vector $\mathbf{g}$ is computed from the original text embedding. The adapted text embedding is obtained by:

$$\tilde{\mathbf{t}} = \mathbf{t} + \mathbf{g} \odot \Delta \mathbf{t}, \quad (1)$$

where $\odot$ denotes element-wise multiplication.

The residual path preserves the original CLAP semantic representation, while the gate vector controls the contribution of the adapted representation in a dimension-wise manner. This highway-style [18] formulation allows the model to adjust the text embedding without substantially perturbing the pretrained semantic space. The resulting enhanced audio-text representation is used for second-level category prediction and also serves as the basis for the neighborhood-based refinement described next.

2.2. KNN-Based Post-Processing

Samples with similar acoustic-semantic representations are likely to have similar class distributions, making their neighbor label distribution useful for ambiguous predictions. As shown in Fig. 1(c), we employ KNN-based post-processing to refine uncertain model predictions using local neighborhood information in the embedding space [11], [12].

After training, we construct a retrieval bank from the training set. For each sample, we construct its retrieval feature by concatenating the

normalized model-enhanced audio representation $\mathbf{h}_a$, the normalized original CLAP audio embedding $\mathbf{a}$, and the normalized original CLAP text embedding $\mathbf{t}$, following the notation marked in Fig. 1(a):

$$\mathbf{u} = \text{Concat}(\text{Norm}(\mathbf{h}_a), \text{Norm}(\mathbf{a}), \text{Norm}(\mathbf{t})), \quad (2)$$

where $\text{Concat}(\cdot)$ denotes feature-dimensional concatenation, and $\text{Norm}(\cdot)$ represents ℓ_2 normalization. The final retrieval representation $\mathbf{z}$ is then obtained by re-normalizing $\mathbf{u}$:

$$\mathbf{z} = \text{Norm}(\mathbf{u}). \quad (3)$$

For a test sample represented by $\mathbf{z}$, the top- K nearest neighbors are retrieved from the bank according to cosine similarity, and the retrieved set is denoted by $\mathcal{N}_K(\mathbf{z})$. The cosine similarity s_i between $\mathbf{z}$ and each neighbor $i \in \mathcal{N}_K(\mathbf{z})$ is then converted into a normalized weight using a temperature-scaled softmax:

$$w_i = \frac{\exp(s_i/\tau)}{\sum_{j \in \mathcal{N}_K(\mathbf{z})} \exp(s_j/\tau)}, \quad (4)$$

where τ controls the sharpness of the neighbor weights. The neighbor label distribution is then obtained by aggregating the retrieved second-level labels:

$$p_{\text{neighbor}}(c) = \sum_{i \in \mathcal{N}_K(\mathbf{z})} w_i \mathbb{I}(y_i = c), \quad (5)$$

where $c \in \mathcal{C}$ denotes a candidate second-level category, $\mathcal{C}$ is the set of second-level sound categories, y_i is the second-level label of the i -th neighbor, and $\mathbb{I}(\cdot)$ is the class indicator function.

After obtaining the neighbor label distribution, we apply confidence-gated fusion to avoid unnecessarily modifying reliable predictions, following the common practice of using prediction confidence to decide whether a model output should be trusted [19], [20]. The original model prediction p_{model} is retained when the confidence is sufficiently high; otherwise, it is interpolated with the neighbor label distribution p_{neighbor} :

$$p_{\text{final}} = \begin{cases} p_{\text{model}}, & \max_{c \in \mathcal{C}} p_{\text{model}}(c) \geq \delta \\ (1 - \alpha)p_{\text{model}} + \alpha p_{\text{neighbor}}, & \text{otherwise} \end{cases}, \quad (6)$$

where δ is the model-confidence threshold, α controls the contribution of the neighbor label distribution, and $p_{\text{model}}(c)$ represents the original model’s predicted probability for category c . Finally, the predicted second-level class is determined by

$$\hat{y} = \arg \max_{c \in \mathcal{C}} p_{\text{final}}(c). \quad (7)$$

2.3. BSD-Grand: Expanded Training Dataset Construction

To improve data diversity while controlling the label noise introduced by external data, we construct an expanded training set, denoted as BSD-Grand. Specifically, we select high-confidence samples from BSD35k-CS [14] and add them to the BSD10k-v1.2 [8], [13] training set using a metadata-cleaning and teacher-voting strategy.

We first construct three metadata variants for BSD10k-v1.2, including raw, v1, and v2 versions. The raw version keeps the original textual metadata unchanged. The v1 version applies basic cleaning to titles, tags, and descriptions, including removing file extensions, redundant separators, HTML tags, URLs and HTML entities. The v2 version further performs category-aware cleaning, where boilerplate and device-related text are removed more aggressively, while sound-discriminative information is preserved for category discrimination.

We then train three teacher models on BSD10k-v1.2, using the raw, v1, and v2 metadata variants separately. The trained teachers are then used to predict the labels of BSD35k-CS samples. For each sample, we compare the teacher predictions with its original label and compute the mean confidence assigned to that label. A sample is regarded as reliable if at least two out of the three teachers predict its original label as the top-1 class and the mean confidence exceeds 0.70, otherwise it is discarded rather than relabeled.

Finally, we apply an uploader-class cap to avoid over-representing dominant uploaders or categories in the expanded training set. For each uploader-class pair, no more than 200 samples are retained. After filtering and capping, 9,573 samples are selected from BSD35k-CS and used only for training.

2.4. Ensemble Strategy

To mitigate the limitations of a single model, we construct an ensemble of complementary models. The ensemble introduces diversity by varying the training objective, acoustic feature representation, and classification head. All candidate models are trained independently and combined only at inference time.

KNN-based distillation: To introduce training-objective diversity, we train a knowledge-distillation-based log-STFT model, denoted as KD-log-STFT, using the KNN-based distillation strategy shown in Fig. 2. A frozen log-STFT teacher is used to extract feature representations for the training samples and construct a teacher feature bank. For each sample, its nearest neighbors are retrieved from this bank, and their labels are aggregated to form a neighbor-aware soft target p_{neighbor} . The student model shares the same architecture as the teacher, but is optimized with a combined objective using both the ground-truth label and the KNN-derived soft target:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{KL}}, \quad (8)$$

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(p_{\text{neighbor}} \parallel p_{\text{student}}), \quad (9)$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss computed against the hard ground-truth labels, $\mathcal{L}_{\text{KL}}$ denotes the Kullback–Leibler divergence between the neighbor-derived soft target p_{neighbor} and the student prediction p_{student} , and λ controls the contribution of the distillation loss. This formulation follows the general idea of knowledge distillation [21], but replaces conventional teacher predictions with a neighbor-aware probability distribution.

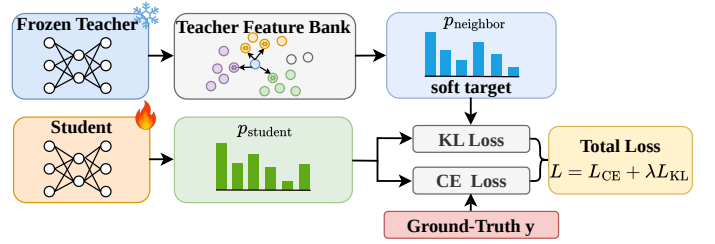


Fig. 2: Illustration of the KNN-based distillation strategy used for training the KD-log-STFT model.

Feature-level diversity: To introduce acoustic-feature diversity, we replace the log-STFT acoustic branch with an alternative log-Mel branch. Consistent with the auxiliary-model setting, this variant does not use the Highway Gate Adapter. The log-Mel branch encodes log-Mel features using a Multi-Dilated Frequency Dynamic Convolution (MDFD) encoder [22], [23], followed by attention pooling. We denote this variant as log-Mel, which provides complementary spectral cues for the ensemble.

Classifier-level diversity: To introduce classifier-level diversity, we modify the baseline¹ classification head in two ways. The first variant, denoted as Flat, adds an additional linear transformation before the prediction. The second variant, based on local classifier per level (LCL) [24], decomposes hierarchical prediction into level-specific classifiers. The LCL-based head used in the ensemble is illustrated in Fig. 3.

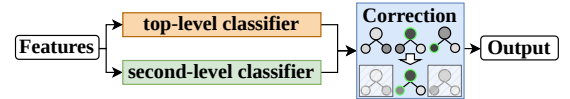


Fig. 3: Illustration of the LCL-based head used as one of the ensemble candidates.

Ensemble inference: At inference time, each candidate model produces a logit vector $\mathbf{o}_m$ for the 23 second-level classes. We combine the models using weighted logit averaging:

$$\mathbf{o}_{\text{ens}} = \sum_{m=1}^M \beta_m \mathbf{o}_m, \quad \sum_{m=1}^M \beta_m = 1, \quad (10)$$

where M denotes the number of ensemble candidates, and β_m denotes the weight of the m -th candidate model. The final class probability p_{ens} is obtained by applying softmax to the ensembled logits:

$$p_{\text{ens}} = \text{Softmax}(\mathbf{o}_{\text{ens}}). \quad (11)$$

All the ensemble weights are selected according to the Hier. F1 performance on a held-out validation set.

3. EXPERIMENTS

3.1. Experimental Setup and Training Details

All audio files are resampled to 16 kHz and converted into fixed-duration 5-second clips, with shorter clips padded to the target length. This fixed-length setting ensures consistent feature dimensions across samples while preserving the main acoustic content for classification. For the acoustic branch, we extract log-STFT features from the processed audio files. Specifically, we use a Fast Fourier Transform (FFT) size of 512, a window length of 400 samples, and a hop length of 160 samples. During training, random cropping [25] and

¹Code: https://github.com/MTG/dc2026_task1_baseline

time masking [26] are applied to the log-STFT features to improve robustness to local temporal variations.

In all experiments, we adopt the AdamW [27] optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 5×10^{-4} . Models are trained for up to 50 epochs with early stopping. Early stopping is triggered if the validation accuracy does not improve by more than 0.1% over 6 consecutive epochs. The loss function is cross-entropy with label smoothing of 0.05.

Furthermore, to ensure fair controlled comparisons, all model variants are evaluated on the same fixed validation and test splits of BSD10k-v1.2. Additional samples from external data are incorporated only into the training set, without modifying the validation or test sets. For KNN-based post-processing, the hyperparameters are selected on the validation set and fixed as $K = 30$, $\alpha = 0.25$, $\tau = 0.05$, and $\delta = 0.85$ for evaluation.

3.2. Contributions of Each Complementary Aspect

To evaluate the contribution of each component in the proposed framework, we conduct component-removal ablation experiments. We use the official audio-text multimodal baseline released for DCASE 2026 Challenge Task 1 as a reference system for comparison. For each ablation variant, we report hierarchical precision (Hier. P), hierarchical recall (Hier. R), and hierarchical F1-score (Hier. F1) before and after applying KNN-based post-processing.

As shown in Table 1, the proposed method achieves the best performance among all variants without KNN-based post-processing, reaching a Hier. F1 of 80.54%. Removing the log-STFT branch decreases the Hier. F1 to 79.97%, demonstrating that the additional time–frequency representation provides complementary acoustic information beyond the CLAP audio embedding. Removing the Highway Gate Adapter reduces the Hier. F1 to 80.05%, indicating that adapting the CLAP text embedding improves its compatibility with the audio representation. Training on BSD10k-v1.2 only, without the selected BSD35k-CS samples, decreases the Hier. F1 from 80.54% to 80.04%, indicating that BSD-Grand data expansion increases training diversity and provides complementary benefits for hierarchical audio classification.

Table 1: Results on the BSD10k held-out test set, w/o KNN and w/ KNN denote without and with KNN-based post-processing, respectively. All models are trained on BSD-Grand unless otherwise specified. All values are percentages.

Framework	KNN	Hier. P $\uparrow$	Hier. R $\uparrow$	Hier. F1 $\uparrow$
Official Baseline	–	79.39	78.25	78.45
Proposed Method	w/o	81.85	79.83	80.54
w/o log-STFT Branch	w/o	81.26	79.31	79.97
w/o Highway Gate Adapter	w/o	81.26	79.45	80.05
w/o BSD35k-CS Data	w/o	81.24	79.38	80.04
Proposed Method	w/	82.23	80.12	80.84
w/o log-STFT Branch	w/	81.89	79.72	80.44
w/o Highway Gate Adapter	w/	82.16	79.85	80.64
w/o BSD35k-CS Data	w/	81.91	79.74	80.51

Table 1 further shows the effect of neighborhood refinement. For all model variants, KNN-based post-processing improves Hier. F1 by incorporating local class distributions from the learned embedding space. This confirms that the neighbor label distribution provides complementary evidence to the model prediction, especially for uncertain samples. The confidence-gated fusion mechanism is also useful because it preserves high-confidence model predictions and mainly refines uncertain cases using neighborhood information.

3.3. Submitted Systems and Evaluation

For the official evaluation of DCASE 2026 Challenge Task 1, we submitted two systems. System 1 is the proposed log-STFT-based single model with the Highway Gate Adapter and KNN-based neighborhood-constrained inference. System 2 is an ensemble system trained once on BSD-Grand. It combines four complementary models, namely KD-log-STFT, log-Mel, Flat, and LCL, which have been introduced in Section 2.4. Among the tested ensemble configurations, this combination achieves the best held-out validation performance, and the final logits are averaged with weights of 0.4, 0.2, 0.3, and 0.1, respectively. Table 2 summarizes the candidate models used in the submitted systems and their performance on the BSD10k held-out test set. Table 3 reports the official evaluation results of the submitted systems.

Table 2: Performance of the official baseline and component models of the submitted systems on the BSD10k-v1.2 held-out test set. All values are percentages.

System	Component Model	Hier. P $\uparrow$	Hier. R $\uparrow$	Hier. F1 $\uparrow$
Official Baseline	–	79.39	78.25	78.45
System 1	–	82.23	80.12	80.84
System 2	KD-log-STFT	81.91	79.92	80.62
	log-Mel	81.14	79.29	79.93
	Flat	81.23	80.30	80.57
	LCL	80.93	79.92	80.20

Table 3: Evaluation results of the submitted systems with official hidden ground-truth labels. The official baseline is the audio-text multimodal baseline released for DCASE 2026 Challenge Task 1. All values are percentages.

System	Hier. P $\uparrow$	Hier. R $\uparrow$	Hier. F1 $\uparrow$
Official Baseline	72.5	69.6	70.0
System 1	81.3	80.2	80.0
System 2	82.5	81.0	81.1

As shown in Table 3, both submitted systems outperform the official audio-text multimodal baseline across all evaluation metrics. System 1 improves the official Hierarchical F1-score from 70.0% to 80.0%, corresponding to an absolute gain of 10.0 percentage points. This result indicates the effectiveness of CLAP representation enhancement and KNN-based neighborhood refinement. The ensemble system further improves the official Hier. F1 from 80.0% to 81.1%, indicating that the component models provide complementary information.

4. CONCLUSION

In this paper, we proposed a CLAP-based heterogeneous audio classification framework for the Broad Sound Taxonomy. The framework integrates a log-STFT acoustic branch, a Highway Gate Adapter, KNN-based post-processing, and BSD-Grand data expansion to improve fine-grained hierarchical classification. Our system shows that high-level audio-text semantic representations can be better adapted to fine-grained BST classification through task-specific audio-text representation enhancement, reliable data expansion, and neighborhood-based label evidence. The best single model, built on the log-STFT architecture with KNN-based post-processing, achieves 80.0% Hierarchical F1-score on the official evaluation set. By further combining KD-log-STFT, log-Mel, Flat, and LCL models with weighted logit averaging, the final ensemble system achieves 81.1% Hierarchical F1-score and ranks second in DCASE 2026 Challenge Task 1. These results suggest that representation enhancement with neighborhood-based refinement is an effective strategy for hierarchical heterogeneous audio classification.

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